

Firm Debt Relief in Financial Downturn

By MATT D'URSO*

I study firm-specific debt relief in a financial crisis through a DSGE model of heterogeneous firms and aggregate uncertainty, calibrated to U.S. data, including the unconditional size distribution, and age-size distribution of firms. A shock to the pledgeability of firms' collateral exacerbates misallocation, causing a crisis. The government tries to optimize debt relief by choosing where to allocate funds. Policy targeting firms with the highest excess return to investment dampens the output trough by 47 percent. Further, if government can only see firm size or age they should target young, or medium size firms, not small firms.

JEL: E22, E32, E62, H32, C60

Can targeting firm debt relief improve stabilization policy in a financial crisis? Around the time of economic downturn, the idea of debt relief, or “bailouts,” becomes more common in political and economic discourse.¹ These policies are usually idiosyncratic, varying in size, scope, and may also allow for changes in market structures. In recent U.S. history, debt relief policies have been implemented during the Great Recession and the COVID-19 pandemic. Motivation for policy during the Great Recession was driven by concerns of larger firms, specifically the “Big 3” in U.S. Auto, laying off workers and a negative wealth effect spreading throughout the economy.² Furthermore, larger firms are responsible for a greater share of aggregate economic activity than smaller firms.³ On the other hand, the

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¹Blau, Brough and Thomas (2013) measure commonality in political discourse with two approximations for political engagement, lobbying and connections maintained.

²General Motors, Ford Motor Company and Chrysler, at the time, were considered the “Big 3” U.S. automobile manufacturers.

³Crouzet and Mehrotra (2020) conclude that cyclical behavior in large firms may have implications for aggregate fluctuations, while small firms' behavior may not. This, along with the positively skewed employment distribution in the United States shown in the U.S. Census Bureau (2022a) (BDS) data may suggest targeting large firms is appropriate.

Paycheck Protection Program (PPP) established during the pandemic was chiefly concerned with smaller businesses, with the main target being firms with fewer than 500 employees.⁴ Rational for targeting smaller firms is based on the fact that, on average, they are likely to grow faster than larger firms on a rate basis, despite weaker capital market access.⁵ Besides identifying which end of the size distribution to focus resources on, it remains to be seen if targeting size at all is the best use of finite funds.

To answer the question, I build a dynamic stochastic general equilibrium model with persistent firm-level heterogeneity, financial frictions, and aggregate uncertainty. The model is calibrated to match key moments in the United States economic data, including the heavily skewed unconditional size distribution of firms, as well as the age-size distribution of young firms, up to the 5th year in their life-cycle. This paper is unique in that, to my knowledge, it is the first of its kind that produces both such distributions. Moreover, the model is overidentified, serving as further validation for the structure and calibration.

The heterogeneity in production creates differences in optimal investment decision rules, while financial frictions can constrain some of these decisions by limiting firms' access to debt markets. The result is a dynamic distribution of firms, which provides a breadth of firm-level variables to target for policy analysis. The aggregate uncertainty in the model is in two dimensions: credit conditions and aggregate productivity. This is a critical feature as the central object that a targeted policy hopes to condition on is the shadow value of a unit of liquidity to a firm. This is a forward-looking expectation that depends on whether aggregate conditions persist, and so directly impacts the efficacy of policy. Further, policymakers assuming they have perfect foresight may misjudge which firms that dollar is worth most to.⁶ I use this model as a quantitative laboratory for policy experiments to study which debt relief policy targets lead to the greatest improvements in aggregate outcomes.

I model financial market frictions by assuming collateralized borrowing. Investment among

⁴The second of three “draws” targeted 300 employees. Other financial targets were also implemented such as limiting net worth and hedge funds.

⁵Evans (1987) and Hall (1987) provide empirical work supporting this claim. Banz (1981) finds that firm size is negatively correlated with risk adjusted returns; much work has been done discussing this correlation and whether or not causality exists (Berk, 1997; Asness et al., 2018). Gertler and Gilchrist (1994) state “While size per se may not be a direct determinant, it is strongly correlated with the primitive factors that do matter,” suggesting policy would be better off targeting smaller firms.

⁶Holmström and Tirole (1998) illustrate the benefits of public liquidity under aggregate uncertainty as the private supply will be insufficient.

firms with insufficient collateral is limited by their inability to borrow, leading to a misallocation of capital. A dynamic fraction of the distribution of firms may not yet have the required capital stock to post sufficient collateral, nor the required retained earnings to reach their efficient investment level, slowing their growth. A shock to firms' access to credit exacerbates the misallocation already present in the model, leading to a deterioration of the aggregate capital stock. The recovery path is a function of the distribution of firms, which is inherently a slow moving object, and if capital requires "time-to-build," the recovery is slowed further, as firms are not able to update their capital immediately. This shock generates a recession with features similar to that of the 2008 Financial Crisis: an initial rise in consumption, a steep fall in investment, and a slow, persistent fall in output.

Financial frictions interact with firm-level productivity, a stochastic process drawn from a bounded Pareto distribution. The asymmetry of the distribution along with imperfect credit markets is what generates the slow growth required to replicate such skewed size distributions.⁷ While the slowed growth seen in young and small firms can also be reproduced in models using the more common log-normal distribution for idiosyncratic productivity, this model delivers something further: many large firms are inefficiently smaller than they should be. Though these firms are considered large by their employment share, a significant portion of them have still not reached their optimal investment level implied by the long right tail of the productivity process.

The paper uses the constrained efficient solution to a planner's problem as a reference for comparison and to theoretically motivate results.⁸ When considering household welfare, optimal debt relief at the onset of a crisis is a policy that decreases and equalizes the highest levels of excess return to capital investment across firms, while internalizing its own pecuniary externality.⁹ The externality means the problem does not have a closed-form solution, but is well-approximated to less than 0.00005 percentage points of household welfare by a closed-form solution that sets this term to 0.¹⁰ This excess return policy directly targets the highest

⁷Shah (2026) is able to replicate the unconditional size distribution with a log-normal productivity process by also modeling product heterogeneity.

⁸The planner can make transfers, but cannot remove capital market frictions.

⁹Excess return to capital investment is defined as the expected lifetime marginal value of investing in capital today, minus its cost.

¹⁰Under the conditions for debt relief policy laid out in the text, this is a price-taking optimal policy. The proof can be found in Appendix A2.

level of value-weighted inefficiency in the economy and lessens the fall of both output and investment by nearly half at their troughs, which is equivalent to a 0.093 percent increase in *permanent* consumption for the household.

Understanding that conditioning on excess return to capital investment may be difficult in practice, I turn to more commonly used size- and age-targeted policy. Using the model’s size distribution targeted to employment size bins found in the [U.S. Census Bureau \(2022a\)](#) (BDS), I divide firms into small, medium, and large size groups.¹¹ Many policies in the United States and abroad focus on financial assistance to small firms.¹² However, I find that debt relief policy that targets medium size firms provides the best results in terms of dampening the fall in output and improving household welfare across size-based policy regimes. Measured at the trough date of the recession under no policy, a medium sized firm targeted policy makes the fall in output and investment over 10 percent less severe, capturing about 20 percent of the welfare gains from the excess return policy.

Financial frictions are more likely to limit small firms’ access to credit in an absolute sense, given the total collateral they can pledge. However, relative to their investment targets, the inefficiency is not as great as for medium-size firms. Idiosyncratic productivity is persistent; small firms have a consequently small target level of capital. Moreover, large firms have exponentially more resources at their disposal, meaning there is less misallocation that policy can correct due to their ability to self finance. Medium size firms continue to grow for a longer period of time, and to a larger size, than small firms, while being more likely to be up against their constraints than large firms.

Age-based policies target young, middle-age, and mature firms (ages 0-5, 6-10, 11+, respectively).¹³ Age is chosen as a policy target because not only is it a readily measurable variable, high output growth and employment growth firms are disproportionately young and young firms increase their debt issuance more than older firms.¹⁴ Among *both* size and age-based policies, targeting young firms provides the largest boost to aggregates in a

¹¹Firm size bins are defined by data from Census Business Dynamics Statistics, the quantitative details of which are discussed in Section III.

¹²[Horvath and Lang \(2021\)](#), [Jo and Senga \(2019\)](#), [Guner, Ventura and Xu \(2008\)](#).

¹³These age bins are also used by [Kim et al. \(2024\)](#). Further, the middle-age bounds are so chosen because the BDS reports exit rates for firms aged 6-10 as a group.

¹⁴[Haltiwanger et al. \(2017\)](#) document this output and employment growth pattern across firms. [Faff et al. \(2016\)](#) find that firm debt exhibits a “hump” shape over the life-cycle.

financially-driven recession, capturing over 28 percent of the welfare gains that the excess return policy does.¹⁵ Though not as surprising as the medium size firm result, this does bring into consideration the impact that size has on fiscal policy, as it is more common to see size as the main eligibility criterion (with small chosen over medium), rather than age.¹⁶ These results, along with data directly from the [U.S. Census Bureau \(2022b\)](#), showing that, controlling for size, young firms make up the largest share of high-growth rate firms, suggest that policy directives may need to be reconsidered.

The benefits of debt relief are not without costs. The household pays a lump-sum tax to fund the policy; they forego present consumption at the onset of the crisis for the potential of an improved recovery path. Since all prices are a function of the marginal utility of consumption, this merits specific consideration as the model is solved non-linearly and equilibrium decision rules are being resolved for all firms at each date. The shape of the endogenously evolving distribution matters for both the direct response to an injection of policy resources, but also the general equilibrium effects of price changes, especially when the shock is not proportional across the distribution, and prices are being forecasted by the household with uncertainty over future states.

The remainder of the paper is prepared as follows. Section I provides a review of the literature. Section II describes the model and provides analysis. Section III begins the quantitative exercise with calibrations. Section IV quantifies measures of efficiency in the model, while section V discusses policy to remedy the effects of a financial crisis. Section VI discusses welfare results and Section VII concludes.

I. Literature Review

There is a large literature on fiscal policy improving macroeconomic outcomes. [Bianchi \(2016\)](#) studies a model of systemic and idiosyncratic debt relief, where systemic policy is a function of only the aggregate state, while idiosyncratic policy takes firm-level variables as arguments, as well. Debt relief is intended to alleviate working capital constraints, ala [Jermann and Quadrini \(2012\)](#). Under an idiosyncratic policy regime, the potential for

¹⁵[Babina et al. \(2019\)](#) find similar employment dynamics between firms of age 5 and firms of ages 4 and 6, suggesting the explicit margin may not need to be so strict.

¹⁶See [Congressional Budget Office \(2020\)](#).

debt relief enters into the firm's decision rules, creating a potential moral hazard problem. Firms are incentivized to over-borrow and may be unable to cover the cash flow mismatch between their outlays and revenue next period. In this setting, systemic policy outperforms idiosyncratic policy.

This paper goes beyond the identical firm framework, bringing more depth to the analysis of idiosyncratic policy. Indeed, I find the opposite result; targeted policy outperforms untargeted policy.¹⁷ Given the non-degenerate distribution of firms and the specific targets tested, most eligible firms are already at (or near) their borrowing limit anyway. Thus even after the policy takes effect, they are not responsible for more debt than before.

In this spirit, I turn to [Buera, Moll and Shin \(2013\)](#). Agents are given the choice to be laborers or entrepreneurs and the government targets high productivity, low wealth agents with a free capital policy. While there are short run benefits of subsidizing productive entrepreneurs, in the long run that productivity will eventually decay. Long run costs accrue from policymakers being unable to adjust the original subsidy policies (for a myriad of potential reasons) and end up funding unproductive entrepreneurs. I depart from [Buera, Moll and Shin](#) in a few ways. First, this work concerns itself with potential stimulus during recession. Policy is intended to counteract the effects of failing credit markets. Although many well-intended policies have inefficient inertia, I study those that exist only for a short period, as in the motivating examples. Moreover, these policies do not distribute free capital. If a firm receives debt relief, the still must pay for investment. Related to this, the interest rate will endogenously change with the aggregate state, rather than being set by a large open economy as in the two aforementioned papers.

[Jo and Senga \(2019\)](#) study fiscal responses to credit market imperfections. As they point out, there are important general equilibrium effects to consider intrinsic in extended access to debt, and therefore capital. Increased factor prices make investment more difficult for ineligible firms. Moreover, these prices effect entry and continuation decisions that are made on extensive margins. The general equilibrium effect may dominate the benefits of policy depending on the idiosyncratic shock process assumed. While their paper focuses on long-run resource allocation through steady state comparisons, this paper will abstract away from

¹⁷See online appendix.

endogenous entry and exit and focus on transitional dynamics.

In this work, policy is conducted by a government, which collects taxes in order to fund debt relief. Here, the government's role is akin to [Woodford \(1990\)](#), by providing a public avenue to (partially) circumvent financial constraints.¹⁸ [Holmström and Tirole \(1998\)](#) expand this idea in the presence of liquidity shocks. [Angeletos, Collard and Dellas \(2023\)](#) solve, non-linearly, for optimal dynamics of tax and public debt in the presence of collateral constraints. The benefit of extended borrowing capacity depends on to whom it is extended. Thus, it is important to consider potential targets to discover if targeted policy, or policy in general, is worth the cost of higher public debt. In order to focus more on the relevant asset distribution and gains from targeting, optimal public debt levels will be left to future work.

Finally, this work leans on literature studying financial frictions, namely collateralized borrowing as in [Khan and Thomas \(2013\)](#). Young firms grow as their ability to borrow rises, bounded by their collateral.¹⁹ Including persistent, yet uncertain, idiosyncratic productivity shocks means optimal capital choices will vary across firms, and change dynamically, creating a richer distribution of firms over excess return. This allows me to explore different dimensions of policy targeting.

II. Model

A. Firms

The model economy is populated with a unit mass of firms that use labor, n , and predetermined capital, k , to produce a homogeneous consumption good through a decreasing returns to scale, twice differentiable, production function, $y = z\varepsilon f(k, n)$. Firm productivity is subject to both aggregate, z , and idiosyncratic, ε , shocks. I assume both z and ε follow their own respective Markov processes: $z \in \mathbf{Z} \equiv \{z_1 \dots z_{N_z}\}$, where $\Pr(z' = z_g \mid z = z_f) = \pi_{f,g}^z$ and $\varepsilon \in \mathbf{E} \equiv \{\varepsilon_1, \dots, \varepsilon_{N_\varepsilon}\}$, where $\Pr(\varepsilon' = \varepsilon_j \mid \varepsilon = \varepsilon_i) = \pi_{i,j}^\varepsilon$. Both $\sum_{g=1}^{N_z} \pi_{f,g}^z = 1$ and $\sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon = 1$. I assume ε is drawn from a bounded Pareto distribution with bounds ε_L and ε_H , and shape η . Each period, there is a ρ_ε probability that firms keep their current ε . With

¹⁸[Woodford \(1990\)](#) studies public debt; since, in this work, taxes are collected as a lump-sum from a representative household that can smooth consumption, in equilibrium, the welfare implications are the same as if the government borrowed the funds first, then repaid them by raising taxes later.

¹⁹See [Albuquerque and Hopenhayn \(2004\)](#) for results driven by limited enforceability and [Bernanke and Gertler \(1989\)](#) for agency costs.

probability $(1 - \rho_\varepsilon)$, they draw a new value from the ergodic Pareto distribution.

In order to fund investment, i , for next period's capital, k' , firms use their revenues and issue a risk-free bond, b' , and b represents their current financial assets.²⁰ They face collateralized borrowing constraints and may not borrow more than a given fraction, ζ , of their current capital stock each period.²¹ Collateralized borrowing is a reduced form method of capturing the health of financial markets. A higher ζ reflects lenders' positive beliefs in the market, thus they are willing to lend more to firms. On the other hand, a lower ζ reflects just the opposite. The parameter $\zeta \in \mathbf{\zeta} \equiv \{\zeta_1, \dots, \zeta_{N_\zeta}\}$ is stochastic and follows its own Markov process, where $\Pr(\zeta' = \zeta_k \mid \zeta = \zeta_h) = \pi_{h,k}^\zeta$ and $\sum_{k=1}^{N_\zeta} \pi_{h,k}^\zeta = 1$.

Aside from collateral constraints, firms also face a lower bound on dividends. The dividends restriction gives the collateral constraint its 'bite.' If a firm cannot afford to finance investment and they cannot provide enough collateral for a loan, this firm may not make up the difference by issuing sufficiently negative dividends.²² This allows the model to capture some of the effects described in the corporate finance literature on the preference of debt financing over equity financing in a tractable manner, while still focusing a quantitatively robust analysis on debt channels.²³

As firms age, their age being denoted by a , their borrowing capacity increases with capital accumulation. In order to prevent the whole economy from outgrowing financial conditions, each period a fraction, $\pi_d(a)$, of firms exogenously exit the market permanently at the end of a period. To reflect the negative correlation between age and exit rates seen in the BDS, $\pi_d(a)$ is a function of firm age. Thus, a given firm, (k, b, ε, a) , is defined by its current capital, $k \in \mathbf{K} \subset \mathbf{R}_+$, debt level, $b \in \mathbf{B} \subset \mathbf{R}$, idiosyncratic shock realization, and age, $a \in \mathbf{A} \subset \mathbf{R}_+$, with an absorbing state $a = a_{max}$ to close the dimension. An exiting firm is replaced by an entrant with a capital stock of k_0 , debt value of b_0 , and idiosyncratic productivity, ε_0 drawn from the ergodic distribution implied by π_{ij}^ε , from a pool of infinite potential entrants to ensure the mass of firms will not change. The resulting entrant rate is π_e .

²⁰The convention used is $b > 0$ represents positive bonds issued (*in debt*) and $b < 0$ represents net savings.

²¹Many papers study future capital (Kiyotaki and Moore, 1997), or earnings-based (Lian and Ma, 2021) collateral. This will not meaningfully change the analysis, subject to a properly calibrated ζ .

²²See Bianchi (2016) as an example of setting a lower bound on dividends as a way to balance tractability with quantitative robustness.

²³Shleifer and Vishny's 1997 research survey and Brav et al.'s 2005 managerial survey show that some firms will choose external capital or even give up positive NPV investments in order to not decrease dividends.

At the start of the period, firms are made aware of their idiosyncratic, ε , and aggregate productivity, z , the financial state, ζ , and if they will exit the market. There is no penalty for exiting the market; firms know if they will exit and, if so, optimize their labor choice given their current capital and productivity, produce output, and settle any outstanding financial obligations, which had an upper bound from last period's collateral constraint. They do not borrow or invest. This timing is crucial, as it insures all debt is risk-free.

Given the structure of the production sector, I now characterize the state of the economy. The distribution of firms over (k, b, ε, a) is described by the probability measure μ defined on the Borel algebra, $\mathcal{S}$, generated by the open subsets of the product space, $\mathbf{S} = \mathbf{K} \times \mathbf{B} \times \mathbf{E} \times \mathbf{A}$. The distribution of firms evolves over time according to the mapping, Γ , from the current aggregate state, $\mu' = \Gamma(z, \zeta, \mu)$. For expositional ease, let the exogenous components of the aggregate state be $s = (z, \zeta)$ and its transition probabilities be $\pi_{l,m}^s = \Pr(s' = (z, \zeta)_m \mid s = (z, \zeta)_l)$, where each $\pi_{l,m}^s$ is derived from the transition probabilities $\pi_{h,k}^\zeta$ and $\pi_{f,g}^z$. Let the shocks realized in the aggregate state s_l be z_f and ζ_h , for each $l = \{1, \dots, N_s\}$, where $N_s = N_z N_\zeta$. Finally, let S represent the entire aggregate state, $S_l = (s_l, \mu)$.²⁴

THE FIRMS' PROBLEM

With the aggregate state described, I now begin to characterize a given firm's problem. Their cash-on-hand is their output, less wage bill, plus undepreciated capital stock, minus debt obligations, with the potential share of debt relieved denoted as, $g(k, b, \varepsilon, a; S)$. Capital depreciates at rate δ and takes 1 period to build. Therefore, cash-on-hand is written as:

$$(1) \quad x(k, b, \varepsilon, a; S) = z\varepsilon F(k, n) - w(S)n + (1 - \delta)k - (1 - g(k, b, \varepsilon, a; S))b$$

and $(1 - g(k, b, \varepsilon, a; S))$ is the fraction of debt the firm is responsible for. This policy structure, where the government retires a fraction of firm debt, is akin to that of [Bianchi \(2016\)](#).

Before making any decisions, a firm's value is then given by $v_0(k, b, \varepsilon, a; S)$, the probability-weighted sum of exiting with cash-on-hand, or continuing on to the next period and receiving

²⁴Hereafter, I suppress the indices for the current exogenous aggregate state l and idiosyncratic productivity, i , except where necessary.

continuation value, $v(k, b, \varepsilon, a; S)$:

$$(2) \quad v_0(k, b, \varepsilon_i, a; S) = \pi_d(a)x(k, b, \varepsilon; S) + (1 - \pi_d(a))v(k, b, \varepsilon, a; S)$$

Once the period starts, continuing firms choose their dividend payments, D , to borrow or save, b' , at the loan discount factor $q(S)$, and capital for next period, k' , while not exceeding their collateral constraint of $b' \leq \zeta k$. Capital accumulation is $k' = i + (1 - \delta)k$. Given the transition conditions of the aggregate and idiosyncratic states, continuation value is:

$$(3) \quad v(k, b, \varepsilon_i; S_l) = \max_{n, k', b', D} \left[D + \sum_{m=1}^{N_s} \pi_{l,m}^s d_m(S_l) \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon v_0(k', b', \varepsilon_j; S'_m) \right]$$

subject to:

$$(4) \quad \begin{aligned} 0 &\leq D \leq x(k, b, \varepsilon_i; S_l) + q(S_l)b' - k' \\ b' &\leq \zeta_l k \\ \mu' &= \Gamma(S_l) \end{aligned}$$

where $d_m(S_l)$ is the relative valuation of dividends next period, conditional on the current and expected future state.

There are no inter-temporal factors of labor demand and the wage bill is paid out of production, meaning all firms with the same (k, ε) have the same labor demand. Thus, the firm's labor decision rule is $g_N(k, \varepsilon; S)$.²⁵ Similarly, a firm's output is $y(k, \varepsilon; S)$. Given $\pi_d(a)$ varies with age, so does the potential valuation of future objects by firms, and that debt relief can condition on age, capital and debt decisions rules require arguments from the full individual state; they are $g_K(k, b, \varepsilon, a; S)$ and $g_B(k, b, \varepsilon, a; S)$, respectively.²⁶

²⁵It is important to state for the proceeding analysis that even though the BDS measures size by employment, k enters the firms' labor decision rule. Financial frictions that limit capital accumulation will also limit firms' growth rate when measured by employment.

²⁶To be described later, "at-risk" firms make choices over k' and b' based on age.

B. Household

There is a representative household endowed with ownership of the firms and a unit of time that may be divided between labor and leisure. The household supplies labor to the firms in exchange for a wage, w , transfers resources intertemporally by purchasing bonds, κ' , and shares, λ' , and chooses consumption, c , to maximize its lifetime value, $W(\lambda, \kappa; S)$. The household discounts the future at rate $\beta \in (0, 1)$ per period and maximizes its periodic utility over consumption and leisure, $U(c, 1 - n)$, subject to their budget constraint. The full household problem is therefore:

$$(5) \quad W(\lambda, \kappa; S) = \max_{c, n^h, \lambda', \kappa'} \left[U(c, 1 - n^h) + \beta \sum_{m=1}^{N_s} \pi_{l,m}^s W(\lambda', \kappa'; S'_m) \right]$$

subject to:

$$(6) \quad \begin{aligned} c + q(S)\kappa' + \int \rho_1(k', b', \varepsilon'; S_l) \lambda' (d[k' \times b' \times \varepsilon' \times a']) &\leq \\ w(S)n^h + \kappa + \int \rho_0(k, b, \varepsilon; S_l) \lambda (d[k \times b \times \varepsilon \times a]) & \\ \mu' = \Gamma(S) & \end{aligned}$$

where $\rho_1(k', b', \varepsilon', a'; S_l)$ is ex-dividend price of a share and $\rho_0(k, b, \varepsilon, a; S_l)$ is the dividend-inclusive value of a share. Their decision rules are: $h^C(\lambda, \kappa; S)$ for consumption, $h^N(\lambda, \kappa; S)$ for labor hours, and $h^\kappa(\lambda, \kappa; S)$ for bonds, and $h^\lambda(k', b', \varepsilon', a', \lambda, \kappa; S)$ for shares in firms with next period values of k' capital, b' debt, and ε' idiosyncratic productivity, of age a .²⁷

C. Government

The role of the government is to conduct debt relief in a crisis. However, the government cannot generate these resources out of nothing; a lump-sum tax is levied on the household in order to fund policy. The benefits of the policy are dependent on resources reaching firms that are financially constrained and face a high marginal productivity of capital to sufficiently

²⁷The household can choose shares of these firm types in equilibrium, since it knows the transition probabilities of the aggregate and idiosyncratic states and the law of large numbers applies. Further, they have access to a complete set of state-contingent claims. These assets are in zero net supply in equilibrium; for simplicity I do not model them here.

compensate the household for the lost consumption. Thus, the government's problem is to maximize household welfare, $\mathcal{W}$, by choice of debt relief targets, subject to their budget T , the firm-level information that is available to them, $\mathcal{I}$, and the path of prices, $\mathcal{P}$. Let the aggregate state at the relief date be denoted as $S_1 = (s_1, \mu_1)$ so that the formal problem is:

$$(7) \quad \max_g \mathcal{W}(g; \mathcal{P})$$

subject to:

$$(8) \quad T(k, b, \varepsilon, a; S_1) = \int_{\mathbf{S}} g(k, b, \varepsilon, a; S_1) b \mu_1(d[k \times b \times \varepsilon \times a])$$

$$(9) \quad g(k, b, \varepsilon, a; S_1) \in [0, \mathbb{1}\{b > 0\}]$$

$$(10) \quad g \text{ is } \mathcal{I}\text{-measurable}$$

$$(11) \quad \mathcal{P} \text{ taken as given}$$

Finally, constraint (9) imposes that this policy must be debt relief. The government cannot target firms that are net savers and cannot relieve more than 100 percent of firm debt.

D. Recursive Equilibrium

A *recursive competitive equilibrium* is a set of functions that solve the firms' and household's problems and clear the asset, labor, and output:

$$(w, q, \{d_m\}_{m=1}^{N_s}, \rho_0, \rho_1, v_0, g_N, g_K, g_B, g_D, W, h^C, h^N, h^\lambda, h^\kappa)$$

(i) v_0 solves (2-3) subject to (4), g_N is the exiting firms' decision rule, and (g_N, g_K, g_B, g_D) are the decision rules for continuing firms

(ii) W solves (5) subject to (6) and $(h^C, h^N, h^\lambda, h^\kappa)$ are the household's decision rules

(iii) Share market clearing:

$$h^\lambda(k', b', \varepsilon_j, a', \lambda, \kappa; S) = \mu'(k', b', \varepsilon_j, a'; S) \text{ for } (k', b', \varepsilon_j, a') \in \mathbf{S}$$

(iv) Labor market clearing: $h^N(\lambda, \kappa; S) = \int_{\mathbf{S}} [g_N(k, \varepsilon; S)] \mu(d[k \times b \times \varepsilon \times a])$

- (v) Goods market clearing: $h^C(\lambda, \kappa; S) = \int_{\mathbf{S}} \left[z\varepsilon F(k, n(k, \varepsilon; S)) - (1 - \pi_d(a))(g_K(k, b, \varepsilon, a; S) - (1 - \delta)k) + \pi_e((1 - \delta)k - k_0) \right] \mu(d[k \times b \times \varepsilon]) - T(k, b, \varepsilon, a; S)$
- (vi) $\forall (A, \varepsilon_j, a + 1) \in \mathcal{S}$ defines Γ , where $\chi(k_0) = \{1 \text{ if } (k_0, b_0) \in A; 0 \text{ otherwise}\}$
- $$\mu'(A, \varepsilon_j, a) = \int_{\{(k, b, \varepsilon_i, a) | (g_K(k, b, \varepsilon_i, a; s, \mu), g_B(k, b, \varepsilon_i, a; s, \mu)) \in A\}} [1 - \pi_d(a)] \pi_{ij} \times$$
- $$\mu(d[k \times b \times \varepsilon_i \times a]) + \pi_e \chi(k_0) H(\varepsilon_j)$$
- (vii) Bond market clearing via Walras's Law: $h^\kappa(\lambda, \kappa; S) = \int_{\mathbf{S}} [g_B(k, b, \varepsilon, a; S)] \mu(d[k \times b \times \varepsilon \times a])$

E. Prices

I begin my analysis by deriving a set of optimality conditions from the household's problem that may be subsumed into the firms' problem. Let C and N represent the optimal household choices of consumption and labor hours, respectively, with C' and N' being the optimal choices next period, when the aggregate state is S' . Then the real wage (12) is the marginal rate of substitution between leisure and consumption. Since the household owns the firms, the firms' stochastic discount factor (13) is the household's marginal rate of substitution between consumption across states. The loan discount factor (14), is the conditional expectation of the marginal rate of substitution between consumption across states.²⁸

$$(12) \quad w(S) = \frac{D_2 U(C, 1 - N)}{D_1 U(C, 1 - N)}$$

$$(13) \quad d(S) = \beta \frac{D_1 U(C'_m, 1 - N'_m)}{D_1 U(C, 1 - N)}$$

$$(14) \quad q(S) = \beta \sum_{m=1}^{N_\varepsilon} \pi_{l,m}^s \frac{D_1 U(C'_m, 1 - N'_m)}{D_1 U(C, 1 - N)}$$

Let $p(S) \equiv D_1 U(C, 1 - N)$ be the household's marginal valuation of output. Since the household owns the firms, I multiply the firms' value (3) by $p(S)$ to price all output in units

²⁸The risk-free interest rate on bonds is: $r = \frac{1}{q(S)} - 1$.

of marginal utility without altering decision rules. Letting $V_0 = p(S)v_0$ and $V = p(S)v$, I rewrite the firms' problem (3), subject to (4), as:

$$(15) \quad V(k, b, \varepsilon_i; S_l) = \max_{k', b', D} \left[p(S)D + \beta \sum_{m=1}^{N_s} \pi_{l,m}^s \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon V_0(k', b', \varepsilon_j; S'_m) \right]$$

F. Allocations

A firm's labor choice is static. It chooses $n^*(k, \varepsilon) = z\varepsilon D_2 F(k, n) = w$. Output is then $y(k, \varepsilon) = z\varepsilon F(k, z\varepsilon D_2 F(k, n))$, which can be used to define cash-on-hand in (1). Turning from static to dynamic decisions, I divide the analysis into three groups of firms: unconstrained, at-risk, and constrained firms.

Unconstrained firms can reach their optimal capital level unimpeded by *any* financial constraint, including future constraints. A unconstrained firm's choice of k' depends only on their productivity, $k^*(\varepsilon) = \operatorname{argmax}_{k'} V(k, b, \varepsilon, a; S)$. These firms continue to invest until $\frac{\beta}{p(S)} (\mathbb{E}_{\pi^s} \mathbb{E}_{\pi^\varepsilon} D_1 V_0(k', b', \varepsilon_j, a'; S'_m)) = 1$. It is useful to define $k^f(\varepsilon)$ as the level of capital that a firm would choose in a world where financial constraints did not exist at all. Tautologically, unconstrained firms choose $k'(\varepsilon) = k^*(\varepsilon) = k^f(\varepsilon)$. The expected marginal discounted (recall $p(S'_m)$ is embedded in V_0) value of an extra unit of capital equals its price, 1 output unit. Applying the [Benveniste and Scheinkman \(1979\)](#) condition, this is:

$$(16) \quad \frac{\beta}{p(S)} \left[\sum_{m=1}^{N_s} \pi_{l,m}^s p(S'_m) \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon \left(z_m \varepsilon_j D_1 F(k^*(\varepsilon), n^*(k^*(\varepsilon), \varepsilon_j)) \right. \right. \\ \left. \left. + (z_m \varepsilon_j D_2 F(k^*(\varepsilon), n^*(k^*(\varepsilon), \varepsilon_j)) - w(S'_m)) D_1 n^*(k^*(\varepsilon), \varepsilon_j) + (1 - \delta) \right) \right] = 1$$

These firms have no excess returns to capital investment. Any debt relief policy will produce no extra value. Another way to see this is that no constraint, nor debt, enter into this optimality condition. Any relief will change neither labor, nor capital, and output will remain the same. Their choice of debt is then $b' \in \left[\frac{D + k^*(\varepsilon) - x(k, b, \varepsilon, a; S)}{q(S)}, \zeta k \right]$, to be discussed further below.

Constrained firms are firms currently bound by the collateral constraint. I take advantage of the monotonically increasing value function in k to simplify the problem. Firms will

borrow until $b' = \zeta k$ and invest such that:

$$(17) \quad k'(k, b, \varepsilon, a) = x(k, b, \varepsilon, a) + q(S)\zeta k - D < k^*(\varepsilon)$$

The strict inequality is the center of the misallocation problem that policy would seek to correct. These firms cannot reach their desired capital level and are unable to hire as much labor or produce as much output next period, compared to an environment without these financial frictions. Moreover, given decreasing returns to scale production, these firms have a higher shadow price on their collateral and higher returns on potential investment than firms without any binding constraints. Again using the properties of the value function and going back to (16), substituting in $k'(k, b, \varepsilon, a)$ for $k^*(\varepsilon)$ returns:

$$(18) \quad \frac{\beta}{p(S)} \sum_{m=1}^{N_s} \pi_{l,m}^s \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon \left[(p(S'_m) + s_{a+1}\lambda'_{m,j}) \left(z_m \varepsilon_j D_1 F(k', n^*(k', \varepsilon_j)) \right. \right. \\ \left. \left. + (z_m \varepsilon_j D_2 F(k', n^*(k', \varepsilon_j)) - w(S'_m)) D_1 n^*(k', \varepsilon_j) + (1 - \delta) \right) \right. \\ \left. \left. + s_{a+1} \zeta'_m \eta'_{m,j} \right] = 1 + \frac{\lambda}{p(S)}$$

There are four terms that did not appear in (16): the multiplier on the current dividend constraint, λ , the multiplier on the future dividend constraint, λ' , the multiplier on the future collateral constraint, η' , and the survival probability s_{a+1} . The survival probability weights future multipliers, which may bind with some probability. This does not matter for unconstrained firms, as by definition, all future multipliers are 0. Future capital influences these multipliers because it has value through its marginal productivity and as collateral. The marginal k' can loosen next period's dividend constraint by increasing the firm's cash, while at the same time loosening next period's collateral constraint by increasing the firm's pledgeable resources. Even if next period conditions allow the firm to escape its collateral constraint, the probability of becoming constrained in the future means the marginal k' still have value to the firm over its cost.

Finally, I move to *at-risk* firms. These firms are unimpeded by their collateral today ($\eta = 0$), however, they face a non-zero probability of binding financial constraints ($\lambda' > 0$) in the future. Due to these potentially binding future constraints, capital investment serves a

dual purpose; it is both productivity and potential collateral next period. These firms make a trade-off between retaining earnings as savings to use in poor future states and investing those earnings in more capital as a buffer stock for insufficient future collateral. This trade-off depends on the expected value of future capital, which is a function of $p(S'_m)$, the exact probability and degree to which future constraints bind, and the survival probability, compared to a bond's known payout of $1/q(S)$ in all states. Thus, the relationship cannot be signed, however, it can be stated that there is no guarantee of $k^*(\varepsilon, a) = k^f(\varepsilon)$.²⁹ The threat of a future binding constraint means that even though the frictionless level of capital is feasible for an at-risk firm, it is unlikely to be their optimal choice. Thus, *the same capital decision rule cannot be assigned to both unconstrained and at-risk firms*. This conclusion is derived in [Lemma 1](#) in the appendix. These potentially binding constraints imply an excess return level above 1, as in (18). The derivation of their decision rules can be found in [Appendix A1](#).

There is still the matter of issuing dividends and the unconstrained firms' debt. I follow the implementation of the *minimum savings policy* as in [Khan and Thomas \(2013\)](#). Firms that are currently borrowing constrained still choose $b' = \zeta k$ and thus dividends are 0. Following from before, at-risk firms have a positive multiplier on next period's dividend non-negativity constraint, so the firm will issue 0 dividends, saving as much as possible, as they value cash more than the household does in equilibrium. Unconstrained firms understand that they can issue some amount of dividends while ensuring that there is no state where they can ever become constrained. They save the minimum amount required to remain unconstrained across all possible future states. This minimum savings policy, and its solution, $B^w(k, \varepsilon; S)$ for these firms are characterized as follows. Let:

$$(19) \quad B^w(k, \varepsilon; S) \equiv \min_{\{\varepsilon_j | \pi_{ij} > 0 \text{ and } S_m | \pi_{lm}^S > 0\}} \tilde{B}(k^*(\varepsilon), \varepsilon_j; S'_m)$$

²⁹Future constraints depend on survival probability, which weighs the present value of future multipliers. Thus, the $k'(\varepsilon, a)$ choice depends on a .

where,

$$(20) \quad \tilde{B}(k, \varepsilon; s, \mu) = z\varepsilon F(k, n^*(k, \varepsilon)) - w(S)n^*(k, \varepsilon) \\ + q(S) \min \left\{ B^w(k, \varepsilon; S), \zeta k \right\} - (k^*(k, \varepsilon) - (1 - \delta)k)$$

and finally, dividends are:

$$(21) \quad D(k, b, \varepsilon, a; S) = x(k, b, \varepsilon; S) - k^*(k, \varepsilon) + q(S) \min \left\{ B^w(k, \varepsilon; S), \zeta k \right\}$$

This is an incredibly conservative approach to firms' dividend decision. Given that idiosyncratic productivity follows a (bounded) Pareto distribution rather than the more commonly used log-normal, the bulk of the ergodic mass concentrates at low productivity, with a long thin spread toward high ε . More large, highly productive, firms faces a non-zero probability of redrawing into the low-productivity region, where the optimal scale, and hence the capacity to service debt without hitting the collateral constraint, falls sharply. This downside risk raises the precautionary value of retained earnings for firms, so a larger slice of the state space chooses to not to pay dividends and instead retain earnings a buffer than would under a log-normal process with the same mean.

It should be noted that a *zero-dividends* policy, $D = 0$ for *all* firms, is also a solution. Constrained firms already take $D = 0$ as given in (17). Firms not constrained by debt market frictions use as much of their retained earnings as needed to invest, and those that need to borrow for optimal investment, will do so to make up the difference. Funds leftover after investment will be used as savings. Given market clearing, the household is indifferent to the firm paying dividends, or retaining any resources leftover after investment as savings; this policy assigns the decision rule to the latter. Therefore, this is an optimal dividend policy for all firms.³⁰

³⁰The (weak) optimality of the zero-dividend policy should not be confused with multiplicity of equilibria, which does not exist in this model. This is a result of the firm's value function being linear in D and b and appropriate discounting of future savings.

III. Calibration and Solution

The model has an annual frequency. The common DRS production function is Cobb-Douglas: $z\varepsilon F(k, n) = z\varepsilon k^\alpha n^\nu$. Entrant firms' capital supply is fixed at χ percent of the steady state aggregate stock: $k_0 = \chi \int k \tilde{\mu}(d[k \times b \times \varepsilon \times a])$. Household utility, $U(c, 1 - n^h) = \ln(c) + \psi(1 - n^h)$, is the result of Hansen (1985) - Rogerson (1988) indivisible labor.

The household discount factor, β is set to 0.962, to imply a 4 percent risk-free real interest rate, as found in Gomme, Ravikumar and Rupert (2011). The parameter governing the marginal utility of leisure, ψ , is set to 2.165, so that in equilibrium, 1/3 of the household's time endowment is spent on market labor. Labor's share of output, ν is set to 0.60 to match Cooley and Prescott (1995). I set the depreciation rate, δ , to imply an average investment-to-capital rate of 0.072, which matches the BEA Fixed Asset Tables data from 1954 - 2006.³¹ χ is set to 0.243, so that the relative employment size of the average entrant is 0.260 of the average firm, matching that in the BDS from 1990-2006. Entrant firm leverage, $\frac{b_0}{k_0}$ is set to 0.41, to match the Ewing Marion Kauffman Foundation (2019) Kauffman Firm Survey. The entrant rate, π_e is set to 0.1068, in order to keep the mass of firms constant at 1, given the vector of exit rates by age, $\pi_d(a)$, taken from the BDS (1990-2006), in Table 1 below. Even though π_e is set via normalization, it is consistent with the BDS entry rate over the same horizon of about 10 percent.

TABLE 1—EXIT HAZARD BY AGE

Age:	1	2	3	4	5	6-10	11+	π_e
Hazard:	0.2478	0.1640	0.1356	0.1174	0.1062	0.0840	0.0655	0.1068

Note: Exit hazards are the fraction of a cohort exiting before the next age, calculated from BDS cohort survival rates from 1990 - 2006. Ages 5 - 10 share a single BDS bin rate, and age 11 is the absorbing age state in the model. The entry rate is not calibrated; it is determined by imposing the mass of firms remains unit every period. It is consistent with the BDS establishment entry rate over the same horizon, roughly 10 percent, providing a check on entry and exit mechanics.

The following parameters are jointly calibrated to match the following moments. The parameter governing capital's share of output, α is set to 0.278 to target a capital output ratio of 2.310. The probability of a firm maintaining its ε draw is 0.9904, to target a standard

³¹U.S. Bureau of Economic Analysis (2026a). To be consistent with earlier work and cited metrics, the intellectual property products is removed from these series and is not considered capital, as per the pre-1999 definition. This does not meaningfully change the calibration.

deviation of investment rates across firms of 0.337, from [Cooper and Haltiwanger \(2006\)](#).³² During an ordinary financial state, the collateral constraint will be $\zeta_o = 0.864$, to target a debt-to-physical asset ratio of 0.269, as found in the Financial Accounts (1990-2006).³³ During a credit crunch, this value changes to $\zeta_l = 0.4907$ in order to facilitate a peak-to-trough decrease in debt of about 32.2 percent. Finally, the model is overidentified; the number of data moments exceeds the number of parameters. The “extra” moment is the population weighted mass of age 0-5 firms, which is a result of the joint calibration and provides further validation of the model. Calibrated parameters are reported on Table 2, along with their closest related data target.

Debt in the model is one-period and secured against the firm’s current capital stock. The empirical counterpart is taken to be loan liabilities of nonfinancial corporate business, relative to fixed capital and nonresidential structures plus equipment. Financial obligations function as loans in the model, so loans are chosen for calibration rather than debt securities. Empirically, bonds and commercial paper are unsecured, issued almost exclusively by the largest rated firms, and underwritten on cash flows rather than collateral ([Lian and Ma, 2021](#)). Further, the credit shock in the paper is disciplined by the contraction in lending seen in the Great Recession, during which bond credit *expanded*. By conditioning on these moments, the steady state and crisis variable are measured against the same stock.³⁴ Deflated by the GDP deflator from [U.S. Bureau of Economic Analysis \(2026b\)](#) data, the same stock falls 32.20 percent from 2008Q4 to 2011Q1.

A. Size Distributions

The model captures the size and age-size distributions in the U.S. through the use of a bounded Pareto distribution for firms’ idiosyncratic shock process, discretized using $N_\epsilon = 15$ values. The discretization process is that of [Jo \(2025\)](#), which calculates the relative employment share of firms in the BDS, defined as small (1-19 employees), medium (20-499), and large (500+). I use those relative employment share measures to define the exact same

³²This is taken from the population of incumbent continuing firms, as in [Cooper and Haltiwanger \(2006\)](#).

³³[Board of Governors of the Federal Reserve System \(2026\)](#). This is taken as the intermediated, non-mortgage credit of nonfinancial corporate business, relative to nonfinancial assets at market value plus liquid financial assets.

³⁴Entrant leverage is measured by total-debt. Entering firms rarely issue bonds or commercial paper, so this and loan measures closely coincide.

TABLE 2—PARAMETERS

Parameter			Target	Data	Model
<i>Panel A: Externally Assigned</i>					
β	discount factor	= 0.962	real interest rate	= 0.040	0.040
ν	labor share	= 0.600	labor share	= 0.600	0.600
δ	depreciation	= 0.072	investment/capital	= 0.072	0.072
$\frac{b_0}{k_0}$	entrant leverage	= 0.410	entrant leverage	= 0.410	0.410
<i>Panel B: Internally Calibrated</i>					
ψ	leisure preference	= 2.165	labor hours	= 0.333	0.333
α	capital share	= 0.278	capital/output	= 2.310	2.320
χ	entrants' share of K	= 0.243	avg. n_0/N	= 0.260	0.260
ζ_o	collateral fraction	= 0.864	debt/capital	= 0.269	0.268
ζ_t	credit crunch	= 0.491	decrease in debt	= 0.322	0.322
<i>Panel C: Productivity Parameters</i>					
ρ_ε	prob. retain ε	= 0.991	s.d. (i/k)	= 0.337	0.337
η	Pareto shape	= 6.200	small firm mass	= 0.885	0.883
ε_H	upper bound ε	= 2.872	medium firm mass	= 0.112	0.111
ε_L	lower bound ε	= 1.000	normalization		
—	—	—	age pop wtd mass	= 0.368	0.370

Note: Panel A shows externally set parameters. Panels B and C lists jointly calibrated parameters, where those in C govern the idiosyncratic shock process. Parameters are listed next to moments they most directly influence. Frequency is annual.

ε_L is normalized to 1 and is not a chosen parameter. The ratio of bounds is the relevant metric, though both are reported for clarity. *Large firm mass* is a residual and therefore not a targeted moment. The model is overidentified and so the *population weighted mass of age 0-5 firms* is a result of the joint calibration.

extensive margins in the model distribution, and sum the mass of firms in each bin to find their population ratio.³⁵ I then check the resulting population share in each bin and compare that to the data. Using the population share of each firm size as calibration targets, I set the upper bound of the Pareto distribution to $\varepsilon_H = 2.872$, and normalize the lower bound to $\varepsilon_L = 1.0$. The shape of the distribution is set to $\eta = 6.20$. The resulting small, medium, and large size bins will be used as eligibility thresholds for size-based policy in Section V.B. The size bin calibration results are reported on Figure 1.

Given χ is chosen to match the average entrant's relative share of employment, the model generates a age-size distribution comparable to what is observed in the U.S. for firms age 0-5,

³⁵Further details in online appendix. Decker, D'Erasmo and Moscoso Boedo (2016) are able to target the firm size distribution with permanent firm level productivity by structuring taste shocks such that the market exposure degree of small firms is limited.

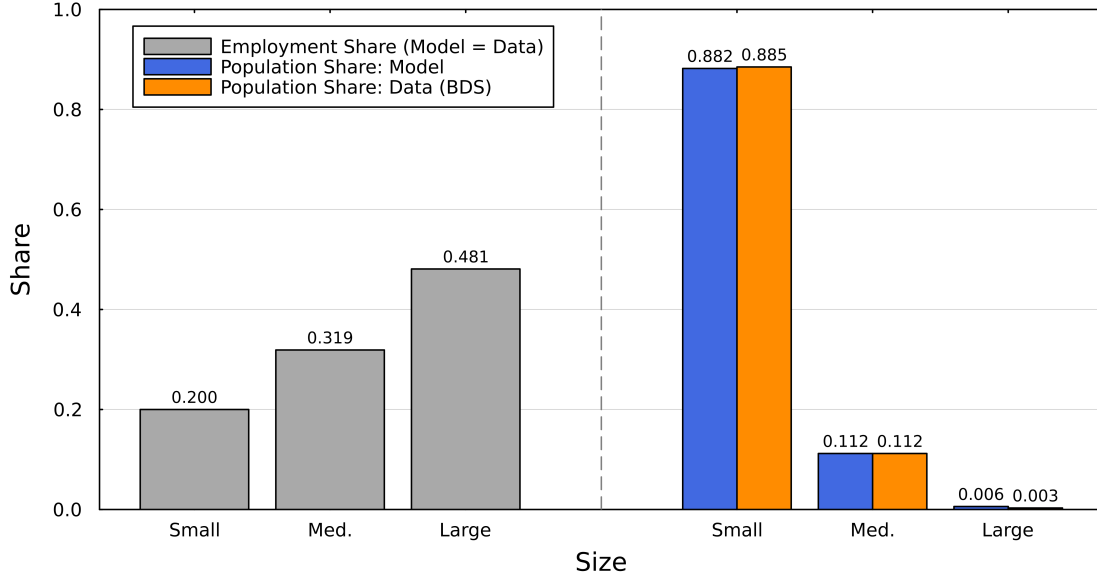


FIGURE 1. UNCONDITIONAL SIZE DISTRIBUTION

Note: Model and data employment and population shares for small, medium, and large firms.

as well as matching the population weighted mean of the age group. Results for ages 1-5 are untargeted and serve as model validation. They are displayed on Table 3. Though the rate of growth measured by employment share is slightly steeper in this model than in the data, it is still monotonic. This outcome is noteworthy given this class of model. Matching the entrant relative size exactly, while trying to slow growth to match empirical moments may lead to endowing entrants with too much capital relative to the aggregate economy. This may lead to surviving entrants deleveraging and shrinking by age 1, and the growth measure losing its monotonicity. This point is even deeper in a model that targets the population weighted average for these firms aged 0-5, as this paper does.

TABLE 3—AGE-SIZE DISTRIBUTION OF FIRMS (BIRTH - YEAR 5)

Age:	Entrant	1	2	3	4	5	Pop. Wtd. Avg.
Model	0.260	0.286	0.346	0.419	0.505	0.599	0.370
Data (BDS)	0.260	0.338	0.378	0.417	0.451	0.477	0.368

Note: Relative employment share of the average firm in each age bin. The model is calibrated to match the relative share of entrants, while the remaining age bins are untargeted.

B. Solution Method

The presence of aggregate uncertainty in the model necessitates a discussion of the solution method. The overall structure follows the [Khan and Thomas \(2013\)](#) implementation of the [Krusell and Smith \(1998\)](#) method. There are some notable differences.

The deterministic steady state is used for calibration and is solved as follows. For a guess on p , calculate the closed form solution of $k^f(\varepsilon)$ for all firms and solve the minimum savings policy. Any firm that returns $D > 0$ is declared unconstrained and their decision rules are solved. The remaining firm's decision rules are solved by simplifying the problem. All other firms choose $D = 0$ and so $b' = (k' - x)/q$; their value function is a function of a single choice variable, k' . Though the grid is rectangular, the choice set is now a line through the space and the value function is concave on that line. Value function iteration, using the [Howard \(1960\)](#) improvement algorithm, and bilinear interpolation are used to solve the value function and decision rules for these firms. Finally, these firms' k' choice is checked against what is feasible given the financial state. If they are constrained by their collateral, they simply follow (17). We can simplify further by realizing the absorbing age state is the only one that feeds into itself. Preceding ages already have a computed $V_0(k', b', \varepsilon', a + 1)$ menu and therefore do not require iteration. Direct aggregation is used and excess demand is checked. Finally, p is updated via [Anderson and Björck \(1973\)](#) and these steps are repeated to convergence.

Moving to the uncertainty over the aggregate state, (z, ζ) , I assume z is a continuous shock following a mean zero AR(1) process in logs, $\log z' = \rho_z \log z + \eta'_z$ with $\eta'_z \sim N(0, \sigma_{\eta_z}^2)$, and the process is discretized to three realizations using the [Rouwenhorst \(1995\)](#) algorithm. Using the empirical estimates of [Khan and Thomas \(2013\)](#), I assign $(\rho_z, \sigma_{\eta_z}) = (0.909, 0.014)$, and assign the values to the transition matrix for the two-state Markov process for ζ , where $p_{o,l}$ are the probabilities of the state persisting, as:³⁶

$$(22) \quad \begin{bmatrix} p_o & 1 - p_o \\ p_l & 1 - p_l \end{bmatrix} = \begin{bmatrix} 0.9765 & 0.0235 \\ 0.3125 & 0.6875 \end{bmatrix}$$

³⁶Their estimates arise from Solow residuals on U.S. real GDP and the share of time advanced economies spend in crisis from 1945 - 2008.

When considering a stochastic equilibrium with aggregate uncertainty over two variables, the previous approach is not feasible. Two refinements are made. First, I assume the household cannot know the full aggregate state. They know s , the the first-moment of the distribution of capital across firms, k_μ , and whether or not credit markets were in a crisis state 1 or 2 periods ago, $(\vartheta_1, \vartheta_2)$.³⁷ Given $(s, k_\mu, \vartheta_1, \vartheta_2)$, the household forecasts (p, k'_μ) . Specifically, I assume the household forecasts $x = \{p, k'_\mu\}$ by:

$$(23) \quad \ln(x) = \beta_0 + \beta_1 \ln(k_\mu) + \beta_2 \vartheta_1 + \beta_3 \vartheta_2$$

Given these forecasted values, equilibrium outcomes are computed, including the market clearing p , for 10,000 periods with the first 1,000 are thrown out. The guess of the forecasting rules is updated and the process repeats to convergence.

The second simplification is in solving the value function on a 1-dimension grid of cash values when determining equilibrium outcomes. First, the value function is solved as above, with the additional aggregate states, once per forecasting rule updating iteration. Next period's value function is interpolated with the forecasted k'_μ , and unconstrained firms still have closed form solutions. Given cash, $V(k', (k' - x)/q, \varepsilon', a')$ is maximized by choice of k' by placing nodes along the curve, bracketing local maxima, maximizing over those sub-intervals, and taking the max over each maximized interval.³⁸ Finally, k' is checked for feasibility and (17) is applied accordingly. This is done to ease the computational burden of searching every (k, b, ε, a) firm each period, ever iteration.³⁹

IV. Quantifying Inefficiency

A. Steady State Inefficiencies

As described in Section II.F, there are inefficiencies present in the model before any aggregate shocks. Financial frictions limit firms' investment and capital accumulation. This

³⁷With unit-mass firms, k_μ is the total amount of capital in the economy. ϑ_1, ϑ_2 are crisis dummies taking on value 1 if $\zeta_{t-1} = \zeta_\ell$ and if $\zeta_{t-2} = \zeta_\ell$, respectively.

³⁸Though the "true" function is concave, the bilinear interpolant does not inherit concavity. This helps safeguard in the even of an interpolation-caused local maximum, so that the algorithm can locate a global solution.

³⁹See online appendix for simulated, HP-filtered business cycle moments comparable to the literature.

creates a wedge between their choice of k' and the optimal k^* . As firms age and grow, they build up a stock of retained earnings to afford more capital investment, relying less on external financing, and thus their collateral. This, however, is a slow process; as evidenced by Table 3, firms in the model and in the U.S. grow slowly when measured by employment relative to the average firm.

As firms move away from their optimal capital choice, their excess return to capital investment grows. From the parameterization of the model, (16) becomes:

$$(24) \quad \frac{\beta}{p(S)} \sum_{m=1}^{N_s} \pi_{l,m}^s \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon \left[(p(S'_m) + s_{a+1} \lambda'_{m,j}) \left(\frac{\alpha}{1-\nu} k'^{\frac{\alpha+\nu-1}{1-\nu}} \right. \right. \\ \left. \left. \left[z_m \varepsilon_j \left(\frac{\nu z_m \varepsilon_j}{w(S'_m)} \right)^{\frac{\nu}{1-\nu}} - w(S'_m) \left(\frac{\nu z_m \varepsilon_j}{w(S'_m)} \right)^{\frac{1}{1-\nu}} \right] + (1-\delta) \right) \right. \\ \left. \left. + s_{a+1} \zeta'_m \eta'_{m,j} \right] = 1 + \frac{\lambda}{p(S)}$$

The further their capital choice is from the optimal level, the more the marginal lifetime benefit of capital investment increases and the Left Hand Side of the equation grows bigger than 1, the marginal cost of a unit of capital.

As expected, excess return declines with age. Firms that have had more time to build retained earnings and their capital stock for collateral are able to invest more. Surprisingly, small firms have the lowest level of excess return, on average. Many small firms are in fact small on purpose; their target capital is smaller than medium and large firms, thus requiring less investment to reach k^* . There is also a general equilibrium effect coming from the labor demand of large firms. The largest 0.6 percent of firms account for 48.1 percent of employment.⁴⁰ These firms drive up the price of labor, disproportionately influencing the optimal capital level of small firms since their k and ε are relatively less influential in Equation (24).

The change in excess return across size groups is not monotonic; medium firms have the largest level on average. These firms are more likely to have a higher idiosyncratic productivity than small firms, increasing their target capital, while also being more effected

⁴⁰In the most extreme case, the largest firm demands 633x the amount of labor as the average firm.

by financial conditions than large firms. Large firms have the most resources at their disposal, but also have the highest k^* to reach. Figure 2 illustrates these results.

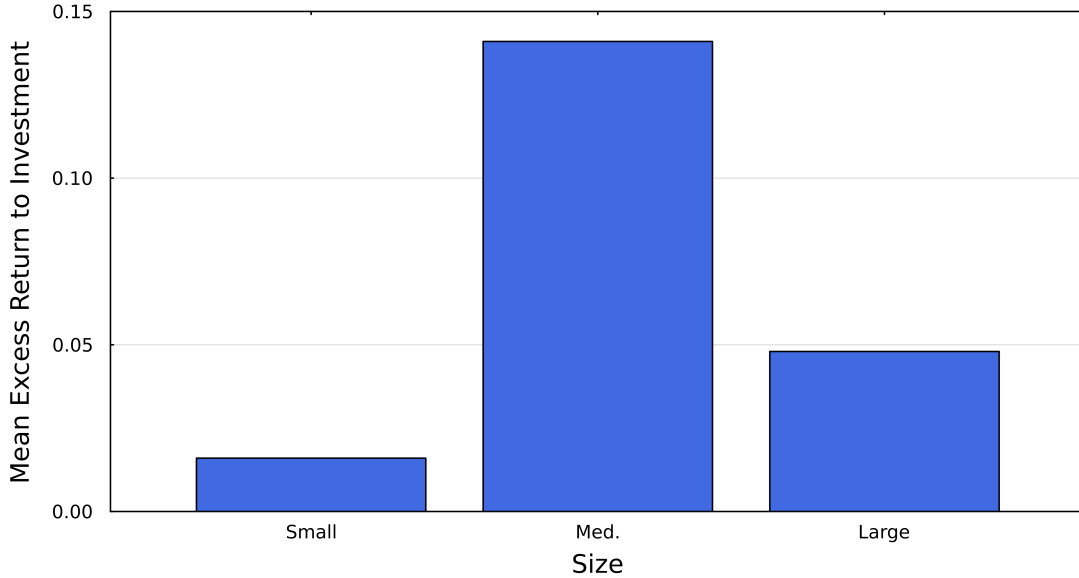


FIGURE 2. AVERAGE EXCESS RETURN TO INVESTMENT BY SIZE

Note: The relationship between excess return to investment and size is non-monotonic. Medium-size firms have the largest average excess return, followed by large, then small.

The relationship between age and excess return is more straightforward. Excess return is monotonically decreasing in age; as firms build up more retained earnings, they are more able to self-finance investment. There is also a compounding effect, especially at low levels of k , which correlates with younger firms, where increases to k loosen constraints. Even if the retained earnings are insufficient to fully self-finance, the loosening constraints increase firms' debt capacity, and thus their investment level. These effects, along with the diminishing returns to capital, create the decreasing and convex shape of Figure 3.

B. Financial Crisis

I now begin the dynamic analysis of this study with a discussion on financial shocks without fiscal policy. The shock in the model is an economy-wide contraction of credit, a fall from ζ_o to ζ_l for four periods, then lasts with a persistence of 31.25 percent per year, as discussed in

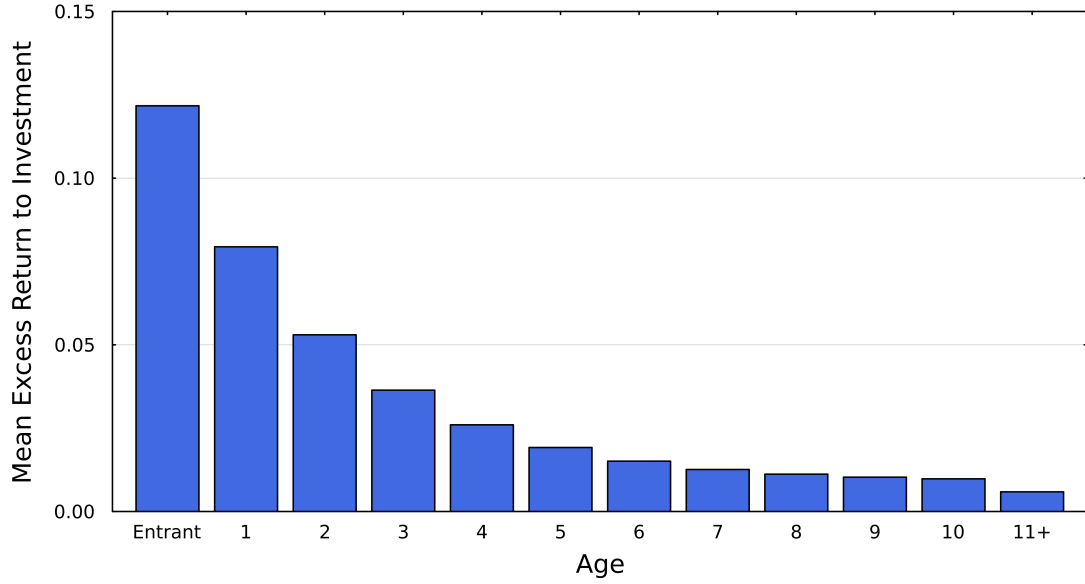


FIGURE 3. AVERAGE EXCESS RETURN TO INVESTMENT BY AGE

Note: The relationship between firm age and average excess return decreases at a decreasing rate.

Section III.⁴¹ Exogenous TFP does not change in this exercise, however, the decision rules of the household are still built under uncertainty; they behave as if z is stochastic during this crisis, as it was during the simulation.

The response to the credit crunch shares defining characteristics with the Great Recession. There is an initial increase in consumption, as once the shock hits the household foresees lower returns to saving. Moreover, the fall in output is persistent as it takes time to rebuild the capital stock after the sharp downturn in investment. The persistent decline in endogenous TFP results from changes in the distribution of firms. Financial requirements have forced the majority of firms to become smaller, increasing misallocation of resources in the economy. Figure 4 displays these dynamics.

V. Debt Relief Policy

The purpose of debt relief in the model is to relax the budget constraint of firms to enable them to afford more investment, offsetting the impact of a credit crunch. The response to the credit conditions laid out is idiosyncratic, implying the impact of different policies will

⁴¹Ivashina and Scharfstein (2010) also find a large decrease in lending at the start of the Great Recession.

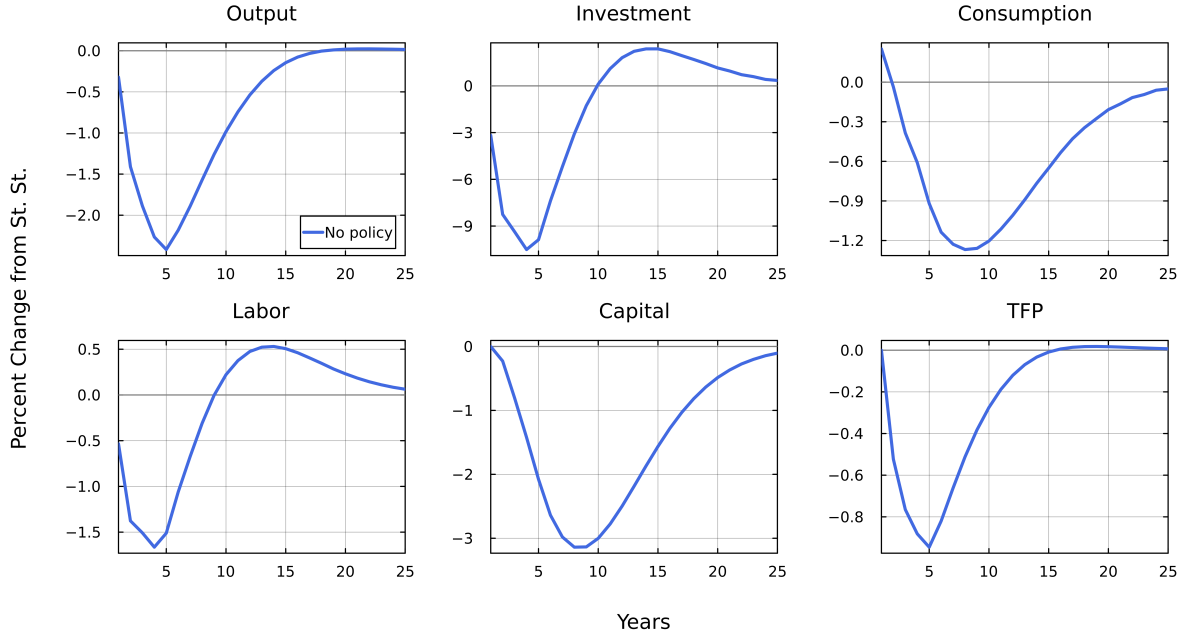


FIGURE 4. FINANCIAL CRISIS WITHOUT DEBT RELIEF

Note: Model response to an unexpected credit shock of ζ_o falling to ζ_l . Shock persists to date 4 and recovers at rate 31.25 percent per year.

be as well. Given a fixed supply of policy funds, policymakers will be concerned with the best return on the resources spent and will focus resources accordingly.

Common across all policy experiments will be the total amount of debt relief paid out, T , to ensure that each policy has the same amount of resources transferred for a fair comparison.⁴² The total resource cost will be 3.5 percent of steady state output, reflecting the fact that the sum of Paycheck Protection Program loan forgiveness was roughly 3.42 percent of U.S. 2020 real GDP (U.S. Small Business Administration, 2022).⁴³ Further, policy is a response to the credit shock described above in Section IV.B and will occur unannounced on impact so it is not something the household can forecast, nor do firm's have any incentive to change their behavior in anticipation of the policy. Thus, the only variation across policy experiments

⁴²It may be the case that endowing the government with more control over model variables will lead to a different T , or possibly removing ζ for select firms. The objective of the exercise is not to study an omnipotent policymaker, but to better understand what optimal targeting looks like given common political constraints.

⁴³This is found by $\frac{F + (P_{20} \div P_{21})}{GDP_{20}}$, where F is total forgiveness and P is core Personal Consumption Expenditures. Since funds were only distributed in 2020 and 2021, this deflation method is sufficient. As of October 2022, 96 percent of all PPP loans have been forgiven (U.S. Small Business Administration, 2022).

is the firm-level information that the government can condition on for targeting. Table 4 summarizes this.

TABLE 4—SCOPE OF INFORMATION & COMMON PARAMETERS

<i>Panel A: Information Scope</i>	
Policy	Scope of $\mathcal{I}$
Excess return	(k, b, ε, a) , continuation
Size	(b, n)
Age	(b, a)
<i>Panel B: Common Parameters</i>	
Policy Cost	0.035 $\bar{Y}$
Credit Shock Dates	1-4
Shock Recovery Rate ($t > 4$)	0.3125
Transfer Date	1

Note: Panel A lists the total information available to policymakers when constructing firm-level targets. Panel B lists parameters that are held constant across all debt-relief policy experiments.

A. Excess Return Policy

The first policy experiment considered directly targets firm level excess return to capital investment. This policy identifies continuing firms with the highest level of excess returns and issues debt relief to them until they match the firms with the second highest level of excess returns. The policy algorithm proceeds in that manner until funds are exhausted.⁴⁴ Taking the equilibrium price sequence under the policy as given, the excess return policy is the optimal debt-relief policy; it is price-taking optimal. In a model such as this, there is no closed-form solution for the globally-optimal policy, once (11) is relaxed, though Appendix A3 approximates the change in welfare from the full solution to be less than 0.00005 percentage points.

The intuition is arbitrage. The household gives up a unit of the consumption good and one unit of debt is retired for a firm. Since the government has full information on the firms, they can ensure that these resources reach continuing firms that will make investments. Multiplying (24) by $p(S)$, the value to the household of the marginal unit of debt retired is

⁴⁴Computationally, one could implement this policy as described in the text. Another way to do so would be to realize that the policy will create a level of excess return that no indebted firm will exceed. A guess of this level implies a total resource cost for the government and this guess may be iterated on until convergence to the exogenously chosen resource cost.

$(p(S) + \lambda)$, strictly greater than the value given up by the household, $p(S)$, when the recipient firm is a continuing constrained or at-risk firm. As resources in the economy are perfectly reversible and the household holds the market, the increase in firm value by $(p(S) + \lambda)$ translates directly to an increase in value of λ for the household. Finally, starting from the highest level of excess return and working down follows a path established by diminishing returns to investment. This ensures that under the excess return policy, the next highest λ is always reached until funds are exhausted.

The excess return policy is strictly beneficial to the household when resources reach a continuing constrained or at-risk firm. This is always the case as the policy naturally establishes a threshold for *eligible debt*. For example, it is theoretically possible for an unconstrained firm to have positive debt. This firm can reach its optimal investment level, so any debt relief resources would just be retained by the firm, or passed on as dividends to shareholders. Though this just gets returned to the household, there is still the opportunity cost of those funds reaching the “wrong” firm. Thus optimal policy would set a threshold of eligible debt such that any relief payments being made will be used for investment by the firm, or should they reach an at-risk firm, be used in expectation, which still is a net benefit for household welfare. This avoids concerns of funds not reaching their intended destination, a concern that has been raised for the Paycheck Protection Program (Li, 2021). This eligible debt threshold derivation can be found in the online appendix, and is:

$$(25) \quad \max \left\{ \min [b, (k^*(\varepsilon) - x(k, b, \varepsilon, a) - q(S)\zeta k)], 0 \right\}$$

This expression makes sure that the firm in question has debt to be relieved in the first place, $b > 0$. Beyond that, the relief policy will only pay out to the firm until they become unconstrained. At that point the relief turns off and the firm will not again become eligible, even if there is remaining $b > 0$. By following the excess returns policy program, this threshold is naturally enforced.

Figure 5 shows the impulse responses of aggregate variables to a credit shock with excess returns-targeted policy implemented. The debt relief leads to major improvements in aggregates during a credit crisis. The output decline at the no policy trough date improves

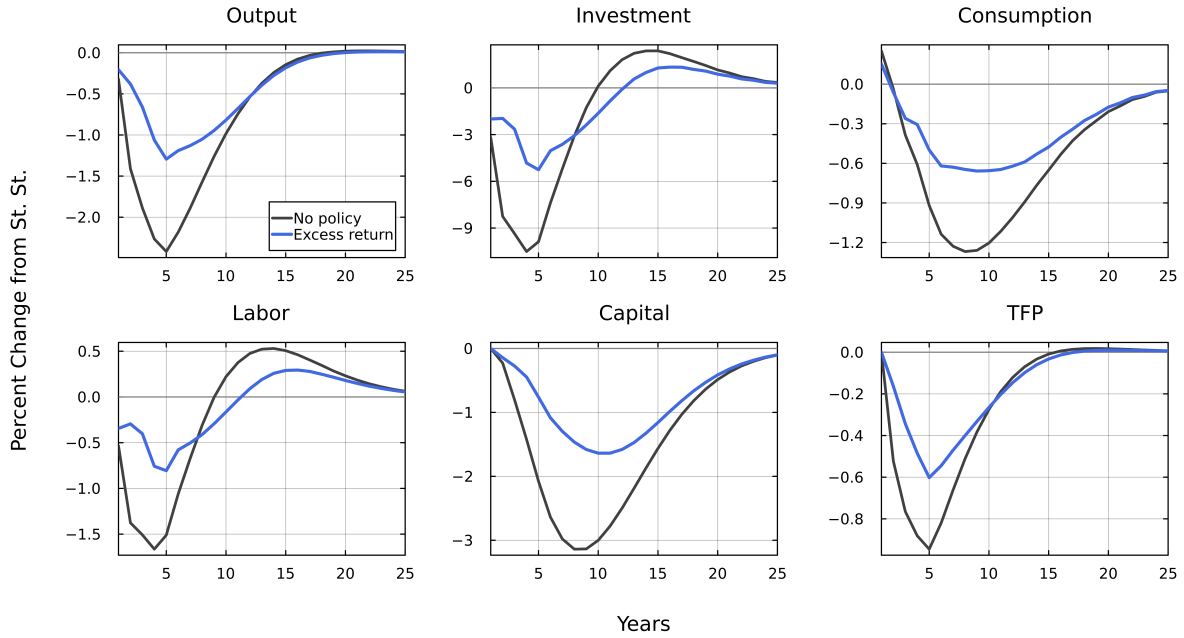


FIGURE 5. RESPONSE TO EXCESS RETURN TO INVESTMENT POLICY

Note: The excess return policy significantly dampens the effect of a crisis across all aggregates.

by 47 percent from a 2.42 percent trough to 1.29. There is a massive initial improvement in endogenous TFP as resources are quickly moved to their most productive location. In fact, there is even a one period rise in labor and investment the date after the policy is implemented, before a second fall. The policy may only last for one period, but its impact is clearly persistent. We also see a sharp initial uptick at date 5. This should be expected as the household understands that this is the first date where it is possible to exit the crisis. This, combined with the fact that the aggregate economy is in a state more conducive to investment post-policy, allows for a relatively sharper improvement at this date.

B. Readily Observable Targets

Identifying excess returns across the state space of an economy may not be feasible for policymakers. To study policies that are more likely to be implemented, I consider two readily observable policy targets: firm size and age.

SIZE-BASED TARGETS

I consider three sized-based policies targeted to small, medium, and large firms. As the total size of each policy will be held constant, and each targeted group is systematically different, the relief per firm must differ across policies. All indebted small firms will have 37.4 percent of their individual debt relieved, indebted medium size firms will have 16.8 percent, and indebted large firms will have 11.0 percent. The following policies shrink the information set of the government leading to two potential sources of opportunity cost: sending a unit of resources to a firm that does not face the highest excess return, or sending it to an exiting firm. That said, since these are such common policy targets, there is value in understanding their macroeconomic implications on a microeconomic level.

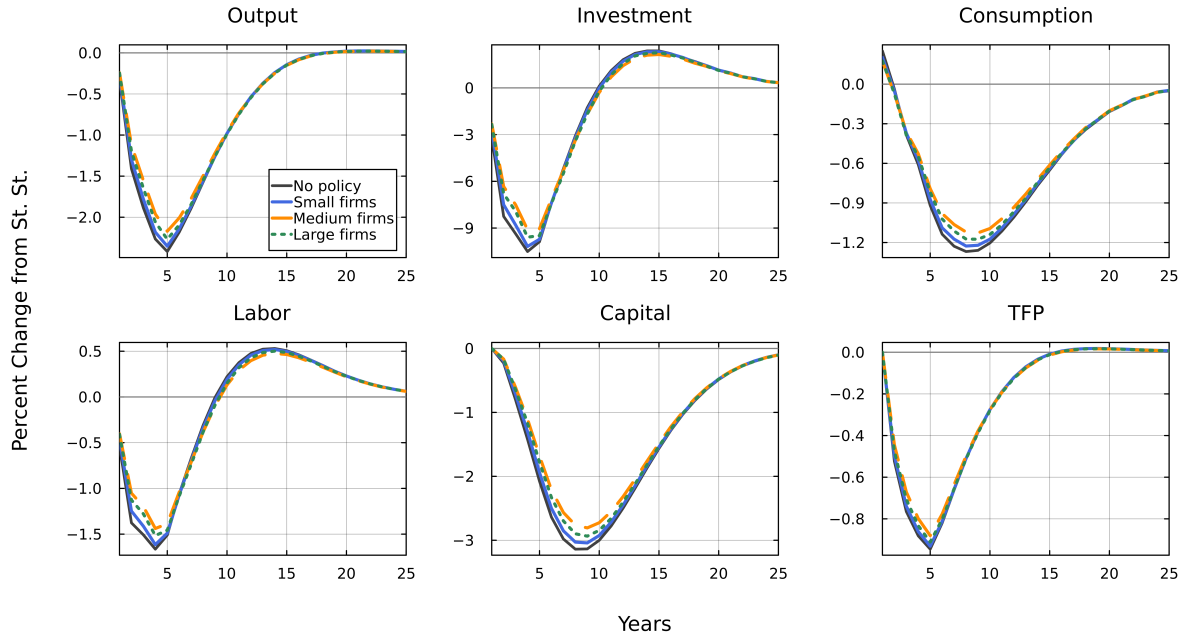


FIGURE 6. RESPONSE TO SIZE-TARGETED POLICY

Note: Across size-targeted policy, aggregates have the highest improvement under medium size policy targets.

Figure 6 reports the response to a credit shock with a size-based policy. Targeting medium firms dampens the fall in aggregates more than targeting small or large firms. Given the persistence of the shock process, small firms' growth potential is much lower than that of medium and large firms. Targeting medium size firms provides resources to firms which are

more likely destined for the long right tail of the distribution, but have yet to outgrow their financial constraints. This is contrasted with large firms that, as a group, already invest at a more efficient level than small and medium size firms, leaving policy with less inefficiency to correct.

The benefit of debt relief is bounded by the investment of receiving firms. Providing debt relief to medium size firms helps ease the burden of financial frictions for firms that will continue to grow far more than small firms, as they have a larger efficient size. Stimulus can then help these firms keep their growth rate higher through the heart of the recession, alleviating more of the persistent effects of the shock. Further, consider the group these firms grow into. Medium size firms are more likely to have a productivity level that, absent financial frictions, would have them remain in the medium size bin, or move to the large size bin. From Figure 1, these firms make up 12 percent of the population mass, but are responsible for 80 percent of aggregate employment and output.⁴⁵ Policy that is able to key in on these firms speeds up their growth process and brings the benefit of their productive capabilities closer to the present date, whereas absent this policy, inefficient investment in a crisis would delay these benefits. Reducing the investment inefficiencies of more productive firms and speeding up the earlier phase of their life-cycle helps maintain a higher aggregate capital stock and endogenous TFP throughout the crisis, thus reducing the magnitude of the output drop at its trough.

AGE-BASED TARGETS

Beyond employment targeting, another proxy for excess returns may be firm age. Given financial frictions, there is a slow growth process to $k^*(\varepsilon)$ as firms age; young firms carry inefficiently low levels of capital. This means that there is a relationship between age and excess returns, and thus potential gains from age-targeted policy.

Age-based policy will target young firms (0-5 years old), middle age firms (6-10 years old), and mature firms (11+ years old). As before, the total size of each policy will be held constant and each group is systematically different; young firms will have 28.6 percent of

⁴⁵It can be shown that for a given firm i , both output share and employment share are $\frac{(\varepsilon_i k_i^\alpha)^{\frac{1}{1-\nu}}}{\int (\varepsilon k^\alpha)^{\frac{1}{1-\nu}} \mu(d[k \times \varepsilon])}$.

their debt paid for, middle age firms will have 20.6 percent of their debt paid, and mature firms will have 10.7 percent.

Figure 7 displays the response to a credit shock under the age-targeted policies. Perhaps, as expected, the young firm policy leads to the best results. This follows from Figure 2 as young firms face higher levels of excess return on investment. Critically, this illustrates the distinction between young and small firms. Young firms may be small because they have not yet had enough time to accumulate retained earnings for investment and are still trying to grow. On the other hand, many small firms are purposefully small. They understand that their productivity is persistent and target a lower level of capital.⁴⁶

Weighing down the benefit to policy targeting young firms is their high exit rate, reported in Table 1, taken from the BDS over 1990-2006. Firms that are receiving policy resources are systematically more likely to exit the economy than those not eligible for the policy. Ideally, policy would lower exit rates and improve results in that regard. On the other hand, one would expect a financial crisis to increase exit rates. It remains to be seen which effect would be stronger, and if one is stronger, will it be enough to significantly change the outcomes presented. I abstract from that discussion and leave it to future work.

VI. Welfare Considerations

The government in this model cannot costlessly conduct policy. Though the impulse response functions generally show improvement, since the household must forego present consumption, I calculate is the percent change in consumption given to the household in the no policy scenario that is required to make the household indifferent to moving to one of the corresponding debt relief scenarios. That is,

$$(26) \quad \sum_{t=0}^{\infty} \beta^t \left[\log \left((1 + \Delta_x) C_t^{\text{np}} \right) - \psi N_t^{\text{np}} \right] = \sum_{t=0}^{\infty} \beta^t \left[\log C_t^x - \psi N_t^x \right]$$

where Δ_x is the percent change in *permanent* consumption required to maintain equality in (26), “np” is no policy, and x is a variable representing one of the policies in the text. The

⁴⁶Consider the case of Microsoft in 1975 and an established local barbershop. The barbershop may have been around for decades, but their capacity is likely nowhere near that of a young Microsoft. With policy resources limited, an omniscient planner would want the bulk of those resources aimed at helping Microsoft grow quickly, especially in a recession.

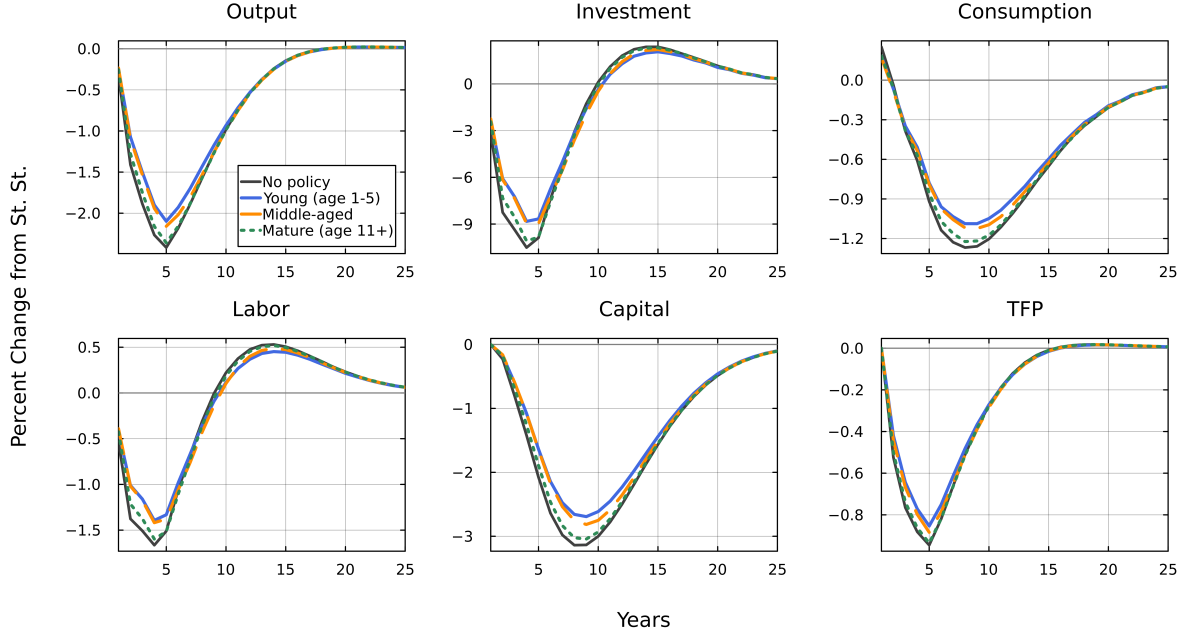


FIGURE 7. RESPONSE TO AGE-TARGETED POLICY

Note: Across both age and size-targeted policy, aggregates have the highest improvement under young-firm policy targets.

welfare ranking is the same minimizing cumulative output loss along the transition path.

TABLE 5—POLICY RANKING BY WELFARE MEASURES

Optimal		Size		Age	
Excess return	0.093%	Medium	0.018%	Young	0.026%
		Large	0.010%	Middle-age	0.016%
		Small	0.005%	Mature	0.004%

Note: Percent change in permanent no-policy consumption to make the household indifferent to each policy.

VII. Concluding Remarks

I develop a dynamic stochastic general equilibrium model where heterogeneous firms face persistent idiosyncratic productivity shocks, financial frictions and aggregate uncertainty. I use this model as a quantitative laboratory to study the stabilization effects of firm-targeted debt relief in a financial crisis. I employ a bounded Pareto distribution to allow the model to capture the heavily skewed size distribution of firms seen in the United States. The

model also produces an age-size distribution of firms that closely matches that of the U.S. Financial frictions limit firm's access to debt, and thus investment, providing a role for debt relief policy in a financial downturn.

Policy that can decrease and equalize the highest levels of excess return to capital investment, the expected lifetime marginal value of capital investment minus its cost, yields the best results of those studied here. It allocates resources to more productive points in the distribution, dramatically cutting the fall in output nearly in half at its no policy trough and drastically increasing endogenous TFP.

Understanding that this may not be a readily measurable policy target, I consider policies based on firm size and age, which have precedent as targets employed by policymakers. I find that policy targeting young firms outperforms other size and age targets, while targeting medium-size firms is the best size-based option. Providing resources to young firms allows them to invest more, speeding up their slow, initial growth phase. Policy targeting medium size firms reduces investment inefficiency in firms with relatively high growth potential, but less resources available to them than large firms, allowing them to grow through the heart of a crisis.

Alternative observables that correlate with excess returns are a potential area for new work. Endogenizing entry and exit decision to allow for debt relief of entrants can be studied as a comparison to the young firms target. I leave these ideas to future research.

APPENDIX

A1. Decision Rules: At-risk Firms

All results across the text hold at any prices that satisfy the stochastic discount factor (14). Firms optimize over labor, choosing $n^*(k, \varepsilon) = z\varepsilon D_2 F(k, n) = w$. Define:

$$\varphi \equiv \frac{\alpha + \nu - 1}{1 - \nu} < 0, \quad c(\varepsilon; S) \equiv z\varepsilon \left(\frac{\nu z \varepsilon}{w(S)} \right)^{\frac{\nu}{1-\nu}} - w(S) \left(\frac{\nu z \varepsilon}{w(S)} \right)^{\frac{1}{1-\nu}}$$

so that operating profits, their derivative w.r.t. k , and cash on hand can be written as:

$$\begin{aligned} \pi(k, \varepsilon; S) &= c(\varepsilon; S) k^{\frac{\alpha}{1-\nu}}, & \pi_k &= \frac{\alpha}{1-\nu} c(\varepsilon; S) k^{\varphi}, \\ x &= \pi(k, \varepsilon; S) + (1 - \delta)k - b \end{aligned}$$

Let the probability of surviving to age $a + 1$ and the expectation operator over aggregate and idiosyncratic states, be:

$$s_{a+1} \equiv 1 - \pi_{a+1}^d \quad \text{and} \quad E_{l,i}[\cdot] \equiv \sum_{m=1}^{N_s} \pi_{l,m}^s \sum_{j=1}^{N_\varepsilon} \pi_{i,j}^\varepsilon[\cdot]$$

$\lambda \geq 0$ is the multiplier on the dividend non-negativity constraint and $\eta \geq 0$ the multiplier on the collateral constraint. Then the firm's problem is

$$(A1) \quad \mathcal{L} = (p(S) + \lambda)(x + q(S)b' - k') + \eta(\zeta k - b') + \beta E_{l,i} V_0(k', b', \varepsilon_j, a+1; S'_m)$$

and the envelope conditions are:

$$(A2) \quad V_k = (p + \lambda)(\pi_k + 1 - \delta) + \eta \zeta \quad \text{and} \quad V_b = -(p + \lambda)$$

Iterating the [Benveniste and Scheinkman \(1979\)](#) result forward and substituting gives:

$$(A3) \quad p(S) + \lambda = \beta E_{l,i} \left[(p(S'_m) + s_{a+1} \lambda') (\pi_k(k', \varepsilon_j; S'_m) + 1 - \delta) + s_{a+1} \zeta'_m \eta' \right]$$

$$(A4) \quad q(S)(p(S) + \lambda) = \beta E_{l,i} [p(S'_m) + s_{a+1} \lambda'] + \eta$$

The collateral constraint is slack for at-risk firms ($\eta = 0$), but they value funds above the household's valuation ($\lambda > 0$), so they pay no dividends. Let the expected values at (k', b') of retained earnings, marginal operating profit, and additional collateral be:

$$\begin{aligned}
 \Lambda(k', b') &\equiv E_{l,i} [p' + s\lambda'], \\
 \mathcal{D}(k', b') &\equiv \beta \frac{\alpha}{1-\nu} E_{l,i} [(p' + s\lambda') c'], \\
 \bar{H}(k', b') &\equiv E_{l,i} [s \zeta' \eta'].
 \end{aligned}
 \tag{A5}$$

Plugging $\eta = 0$ into (A4), substituting into (A3), and defining b'_{ar} as their bond choice:

$$\begin{aligned}
 k^* &= \left[\frac{\left(\frac{\beta}{q(S)} - \beta(1-\delta) \right) \Lambda(k^*, b'_{ar}) - \beta \bar{H}(k^*, b'_{ar})}{\mathcal{D}(k^*, b'_{ar})} \right]^{1/\varphi}, \\
 b'_{ar} &= \frac{k^* - x}{q(S)}, \quad D = 0,
 \end{aligned}
 \tag{A6}$$

The presence of λ' and η' moves the firm's target capital away from the frictionless benchmark, k^f . There are two effects on the sign of this relation, one that pushes k^* above k^f , and one that varies with the state.

LEMMA 1 (Relation of k^* and k^f for At-risk Firms): *For an at-risk firm with state $(k, b, \varepsilon_i, a; S_l)$; let (k^*, b'_{ar}) solve (A6) and k^f solve (16) at the same (ε_i, S_l) and prices, with continuation multipliers evaluated at (k^*, b'_{ar}) ; their decision rule can not be ex-ante pinned to that of an unconstrained firm. Further, $k^* > k^f$, if $\eta' > 0$ with positive probability, or if there is any chance the collateral constraint will bind tomorrow.*

PROOF:

If $\eta = 0$, for k^* and k^f , (A4) gives $p + \lambda = \beta\Lambda/q$ and substituting into (A3), dividing by $\beta\Lambda$, and using $\pi_k = \frac{\alpha}{1-\nu} c' k^\varphi$ gives:

$$\frac{\alpha}{1-\nu} (k^*)^\varphi E^\Lambda[c'] = a_q - \frac{\bar{H}}{\Lambda} \quad \text{and} \quad \frac{\alpha}{1-\nu} (k^f)^\varphi E^p[c'] = a_q$$

Dividing gives:

$$\left(\frac{k^*}{k^f}\right)^\varphi = \left(1 - \frac{\bar{H}}{a_q \Lambda}\right) \frac{E^p[c']}{E^\Lambda[c']}$$

and $\varphi < 0$ results in (A7). The claim follows as $\bar{H} \geq 0$, and $\bar{H} > 0$ when $\eta' > 0$. ■

Define $c' \equiv c(\varepsilon_j; S'_m)$, $a_q \equiv 1/q - (1 - \delta) > 0$, let E^Λ and:

$$E^\Lambda[c'] \equiv \frac{E_{l,i}[(p' + s\lambda') c']}{E_{l,i}[p' + s\lambda']}, \quad E^p[c'] \equiv \frac{E_{l,i}[p' c']}{E_l[p']}. \quad \text{Then,}$$

$$(A7) \quad k^* \geq k^f \quad \Longleftrightarrow \quad E^\Lambda[c'] \geq \left(1 - \frac{\bar{H}}{a_q \Lambda}\right) E^p[c'].$$

There are two forces in (A7). The first always pushes k^* above k^f ; capital has value as collateral so in any case where the firm faces a positive probability of being constrained, the threshold is strictly less than $E^p[c']$. In this case, the firm wants to have more capital as a collateral buffer.

The second force depends on future profits. The firm trades off capital and bonds, where capital pays $\pi_k + 1 - \delta$, which varies with the aggregate state, and thus $(p' + s\lambda')$, and bonds pay $1/q$ in every state. In profitable states where funds are scarce, such as when a higher ε_j raises both c' and λ' , then $E^\Lambda[c'] \geq E^p[c']$. However, the opposite may occur in a crisis where $w = \psi C$ increases at the same time as λ' , making bonds more attractive. Regardless, the claim holds: even though it is affordable, the frictionless capital level is not always the firm's optimal choice.

COROLLARY 1 (Forward-looking Collateral Constraints): *Under a forward-looking constraint, $b' \leq \theta k'$, at-risk and unconstrained firms still have different decision rules, despite no buffer value in investing in more k' today as b'' is secured by k'' , rather than by k' .*

PROOF:

Under this timing, $\eta = 0$ and $D = 0$. The first-order condition is:

$$E_{l,i}[(p' + s\lambda')(\pi_k(k', \varepsilon_j; S'_m) + 1 - \delta - 1/q)] = 0$$

which k^f solves IFF $E^A[c'] = E^p[c']$ in (A7). By definition, this equality will not hold if there is at least 1 state where the firm will no longer be at-risk.

A2. Price-Taking Optimality of the Excess-Return Policy

Relief reaches indebted, continuing firms $i = 1, \dots, N$. A relief schedule is $r \in \mathbb{R}^N$ with $r_i = g_i b_i$, and given $0 < T < \sum_i \mu_i b_i$, the feasible set is:

$$(A8) \quad \mathcal{R}(T) \equiv \left\{ r \in \mathbb{R}^N : 0 \leq r_i \leq b_i, \sum_i \mu_i r_i = T \right\}.$$

Let $\Phi_i(r) \equiv V(k_i, (b_i - r), \varepsilon_i, a_i; S_1)$. $\mathcal{W}$ is strictly increasing in wealth, which policy changes only through recipient firms, so the policy schedules can be ranked by

$$(A9) \quad \mathcal{V}(r) \equiv \sum_{i=1}^N \mu_i \Phi_i(r_i).$$

The payout is linear in D , the firm's constraints are linear in $(b; D, k', b')$, and operating profits are concave in k , so the feasible set is convex, and V and V_0 are concave in (k, b) . (A2) gives $-V_b = p + \lambda = p(1 + \text{ER})$, one-sided at kinks. Each Φ_i is therefore continuous, concave, and strictly increasing on $[0, b_i]$, with $\Phi'_{i,\pm}(r) = p(1 + \text{ER}_{i,\pm}(r))$, where $\text{ER}_{i,\pm}(r) \geq 0$ is firm i 's excess return after relief.

A schedule $r^* \in \mathcal{R}(T)$ maximizes welfare IFF there is a $\overline{\text{ER}} \geq 0$ such that:

$$(A10) \quad \begin{cases} \text{ER}_{i,+}(r_i^*) \leq \overline{\text{ER}} \leq \text{ER}_{i,-}(r_i^*), & 0 < r_i^* < b_i, \\ \text{ER}_{i,+}(0) \leq \overline{\text{ER}}, & r_i^* = 0, \\ \text{ER}_{i,-}(b_i) \geq \overline{\text{ER}}, & r_i^* = b_i. \end{cases}$$

Since $\mathcal{W}$ is strictly increasing in $\mathcal{V}$, and r^* maximizes $\mathcal{V}$ on $\mathcal{R}(T)$. $\mathcal{V}$ is continuous and concave and $\mathcal{R}(T)$ is a nonempty compact polytope, so a maximizer exists. The Karush-Kuhn-Tucker conditions are necessary and sufficient. Letting $\theta \equiv p(1 + \overline{\text{ER}})$ be the multiplier on the budget constraint, $\Phi'_{i,+}(r_i^*) \leq \theta \leq \Phi'_{i,-}(r_i^*)$, with only the first inequality at $r_i^* = 0$ and only the second at $r_i^* = b_i$, which is (A10). Under the calibration, $\overline{\text{ER}} = 0.2160$.

A3. *Approximation of Full Optimality*

Let $\Delta\mathcal{W} \equiv \mathcal{W}(r^*) - \mathcal{W}(0)$ be the welfare gain of the policy over no policy. The gain from the price-taking policy is the integral along the relief path:

$$(A11) \quad \mathcal{G} \equiv p \sum_i \mu_i \int_0^{r_i^*} ER_i(u) du$$

so $\Delta\mathcal{W} - \mathcal{G}$ is the pecuniary term to the second-order. Under the calibration, the externality has a value of -1.02×10^{-4} ; a negative externality to the household, as the initial fall in consumption raises p_1 while lowering w_1 and q_1 .

Policy works through creating K' . The household forecasts K_2 , which is the vector through which the path of future aggregates (and thus prices) will move. To approximate the externality per value-unit of capital created, Let:

$$(A12) \quad \hat{\gamma} \equiv \frac{\Delta\mathcal{W} - \mathcal{G}}{p \Delta K_2}$$

where $\Delta K_2 = \sum_i \mu_i [k'_i(x_i + r_i^*) - k'_i(x_i)]$. Under the calibration, $\hat{\gamma} = -0.0024$. For each $\gamma \in \{0, \hat{\gamma}/3, \hat{\gamma}, 3\hat{\gamma}\}$ the policy is re-solved with $ER_i + \gamma \text{MPI}_i$. Though $\overline{ER}$ changes slightly, dropping about 2.5 percent, welfare changes by less than *0.00005 percentage points*.

A4. Accuracy

TABLE A1—CONDITIONAL FORECASTING RULES

Aggregate Capital: k'_μ							
obs.	β_0	β_1	β_2	β_3	S.E.	adj. R^2	max. error
<i>Panel A: conditional on $s_1 = (z_1, \zeta_o)$</i>							
2006	0.10918 (0.00032)	0.80306 (0.00055)	-0.00404 (0.00014)	-0.00361 (0.00011)	0.00085	0.99908	0.00495
<i>Panel B: conditional on $s_2 = (z_1, \zeta_\ell)$</i>							
0199	0.10344 (0.00156)	0.80956 (0.00264)	-0.00424 (0.00027)	-0.00345 (0.00025)	0.00132	0.99835	0.00368
<i>Panel C: conditional on $s_3 = (z_2, \zeta_o)$</i>							
4170	0.13041 (0.00027)	0.79897 (0.00041)	-0.00430 (0.00011)	-0.00363 (0.00008)	0.00092	0.99892	0.00594
<i>Panel D: conditional on $s_4 = (z_2, \zeta_\ell)$</i>							
0328	0.12393 (0.00157)	0.80576 (0.00241)	-0.00426 (0.00024)	-0.00355 (0.00022)	0.00158	0.99740	0.00563
<i>Panel E: conditional on $s_5 = (z_3, \zeta_o)$</i>							
2143	0.15184 (0.00036)	0.79573 (0.00050)	-0.00459 (0.00013)	-0.00351 (0.00011)	0.00078	0.99917	0.00517
<i>Panel F: conditional on $s_6 = (z_3, \zeta_\ell)$</i>							
0154	0.14764 (0.00180)	0.79848 (0.00253)	-0.00426 (0.00027)	-0.00359 (0.00025)	0.00123	0.99857	0.00335
Output Valuation: p							
obs.	β_0	β_1	β_2	β_3	S.E.	adj. R^2	max. error
2006	0.68115 (0.00008)	-0.41873 (0.00014)	0.00142 (0.00004)	0.00054 (0.00003)	0.00022	0.99978	0.00136
<i>Panel B: conditional on $s_2 = (z_1, \zeta_\ell)$</i>							
0199	0.67972 (0.00039)	-0.41974 (0.00065)	0.00153 (0.00007)	0.00069 (0.00006)	0.00033	0.99961	0.00091
<i>Panel C: conditional on $s_3 = (z_2, \zeta_o)$</i>							
4170	0.64573 (0.00007)	-0.41342 (0.00011)	0.00144 (0.00003)	0.00052 (0.00002)	0.00024	0.99973	0.00154
<i>Panel D: conditional on $s_4 = (z_2, \zeta_\ell)$</i>							
0328	0.64397 (0.00039)	-0.41402 (0.00060)	0.00152 (0.00006)	0.00069 (0.00006)	0.00039	0.99937	0.00145
<i>Panel E: conditional on $s_5 = (z_3, \zeta_o)$</i>							
2143	0.60996 (0.00009)	-0.40821 (0.00013)	0.00144 (0.00004)	0.00052 (0.00003)	0.00021	0.99978	0.00137
<i>Panel F: conditional on $s_6 = (z_3, \zeta_\ell)$</i>							
0154	0.60822 (0.00040)	-0.40892 (0.00057)	0.00150 (0.00006)	0.00068 (0.00006)	0.00028	0.99972	0.00077

Note: Regressions take the form of $x = \beta_0 + \beta_1 \ln(k_\mu) + \beta_2 \vartheta_1 + \beta_3 \vartheta_2$ where $x = \{\ln(k'_\mu), \ln(p)\}$ and ϑ_1, ϑ_2 are crisis dummies taking on value 1 if $\zeta_{t-1} = \zeta_\ell$ and if $\zeta_{t-2} = \zeta_\ell$, respectively. Standard errors in parentheses. [Den Haan \(2010\)](#) errors found by log difference: $e_{\max}K = 0.0160$, $e_{\text{mean}}K = 0.0021$, $e_{\max}p = 0.0015$, and $e_{\text{mean}}p = 0.0002$. These values are all below those of the [Krusell and Smith \(1998\)](#) stochastic- β model.

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